

Ka-Kyung Kim Ph.D.

I design and run the genomics that carries a specimen through to a clinical decision.

Chief Researcher, Diagnosis Division, Cytogen, CTC-based liquid biopsy · Seoul, Korea

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Summary

I have worked with genome, transcriptome, single-cell, and clinical multi-omics data for over twenty years. The analysis carries through to reports and services that hospitals and the business actually use.

Focus Areas

Multi-platform NGS and multi-omics analysis, including scRNA-seq
Clinical trial data analysis (Ph1-3) and regulatory frameworks (GCLP/CAP/ISO)
End-to-end pipeline development with automation and QC standardization
AI/LLM-driven variant interpretation and analysis workflows

Professional Experience

2025-present	Cytogen Chief Researcher, Diagnosis Division Leads CTC diagnostics, scRNA-seq pipeline development, clinical report automation, and GCLP quality systems; first to define the diagnostic service portfolio and its pricing, with the market research behind it
2021-2024	Theragen Bio Director · Clinical Analysis Team / Bioinformatics Led Ph1-3 oncology trial analysis, obtained new GCLP accreditation and designed its data management scheme, analysis lead for the national bio-big-data pilot
2018-2021	Macrogen Deputy GM · Business Development / Clinical Analysis Led K-MASTER genomic analysis, BRCA software with CE-IVD documentation, and liquid-biopsy commercialization strategy
2015-2018	Yonsei University, Biomedical Systems Informatics Postdoctoral Researcher Multi-omics disease variant discovery and non-coding eQTL translational research across gastric cancer PDX, canine mammary tumors, pediatric allergy, and neurological disease
2012-2015	SAIHST, Sungkyunkwan University Postdoctoral Researcher · Molecular Informatics & Genomics Built the miRTCat database; NRF Research Fellow studying disease variants at non-canonical microRNA target sites
2007-2012	Samsung Biomedical Research Institute Researcher · Human Genomics Korean HapMap and KARE national cohort analysis, pharmacogenomics, and genotype imputation for the Korean population
2004-2007	Korea National Institute of Health, Center for Genome Science Research Technician Disease-associated variant discovery in the Korean population; built and operated the SNP database

Selected Programs

Cytogen	CTC-based liquid biopsy diagnostics Built and commercialized the end-to-end operating system for circulating-tumor-cell diagnostics: assay concept, pipeline, clinical report, and customer technical support.
Theragen Bio	Immuno-oncology clinical trial analysis, Ph1-3 Led oncology trial analysis for three pharmaceutical companies.
Macrogen	K-MASTER precision oncology genomics Led genomic analysis for a national precision-oncology platform.
Macrogen	BRCA1/2 variant analysis software, CE-IVD documentation Directed development of software spanning variant calling, clinical interpretation, and automated report generation, together with CE-IVD-grade documentation and performance evaluation.
2021-2024	Quality and regulatory systems Obtained new GCLP accreditation for a clinical specimen analysis laboratory, and established recurring certification cycles — CAP proficiency testing twice a year and ISO27001 annually.

Selected Publications

2019	Whole-exome and whole-transcriptome sequencing of canine mammary gland tumors First author Kim KK, Seung BJ, ... Kim S. Scientific Data 6(1):147
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2013	miRTCat: a comprehensive map of human and mouse microRNA target sites including non-canonical nucleation bulges First author Kim KK, Ham J, Chi SW. Bioinformatics 29(15):1898-1899
2018	Palmitate and minimally-modified low-density lipoprotein cooperatively promote inflammatory responses in macrophages Co-first author Ann SJ†, Kim KK†, ... Miller YI. PLoS One 13:e0193649
2020	Cross-species oncogenic signatures of breast cancer in canine mammary tumors Kim TM, ... Kim KK, ... Kim S. Nature Communications 11:3616

The full list of 20+ papers is on Google Scholar.

Built

In-house · LLM	LLM-based genomic Q&A and clinical report automation An LLM system that answers from test results and source documents (RAG), together with clinical report automation.
Open source · LLM	VeriVar A somatic variant interpretation pipeline.
Multi-agent research	AutoBioX A multi-agent research harness that splits data preparation, prediction, verification, evidence search, and manuscript drafting.
AI agent · Hackathon	Vital STORM An agent that grades key health-screening values on a five-level scale and routes to follow-up examinations, answering from four national health-screening manuals through retrieval (RAG).
Open source · Agent harness	econ-radar An agent harness that assembles and publishes a daily economic digest weighted toward pharma, bio, and AI.

Talks

2026.07	The AutoBioX multi-agent pipeline PseudoCon 2026 · talk
2026.01	Automated Clinical Reporting for Oncology NGS: Dual-Mode LLM Pipeline ABDD Summit 2026 at Stanford · i-Talk and poster
2025.11	Single-cell QC standardization and an scRNA-seq × immunofluorescence pipeline Future of Medicine Symposium: At the Crossroads of Computational Pathology and Spatial Biology · lightning talk
2024.10	Poster presentation, ABDD Summit Awarded Best Poster Presentation
2022.04	Cancer: a history of the emperor of all maladies 1st K-BioX Global Mentoring EXPO · invited talk

Awards

2025	Cheiron ResearchThon — Creative Academic Impact Award · liquid-biopsy multi-omics model for tumor microenvironment prediction	2021	Medical Informatics Advanced Course, Encouragement Award · Asan Medical Center
2024	ABDD Summit, Best Poster Presentation Award · Merit Awards	2019	Korea Clinical Datathon, Grand Prize · sepsis prediction, KoNECT
		2011	KSMCB Annual Meeting, Outstanding Poster Award

Education

1996-2000	Dongguk University B.S. Computer Science	2007-2011	Sungkyunkwan University Ph.D. Molecular Medicine
2002-2004	Korea University M.S. Life Science	2021-2023	Korea National Open University B.S. Health & Environment, Business (double major)

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